An Explainable Attention Network for Fine-Grained Ship Classification Using Remote-Sensing Images

Wei Xiong, Zhenyu Xiong, and Yaqi Cui

***Abstract*— Advances in space-based ocean surveillance systems have improved the detection of objects from high-quality remote-sensing big data. Previous studies mainly focused on finding and recognizing objects based on deep learning and statistical frameworks. Studies have not fully explored the transparent and reasonable decision-making process for the final predicted results, which is vital for civil and military applications. An explainable attention network for fine-grained ship image classification is proposed in the present study to bridge this gap. The present study seeks to increase attention to objects' discriminative parts and explore intrinsic relationships between multiple attention parts and predicted outcomes. Several causal multi-attention maps are generated by combining the multi-head attention mechanism and structural causal model. The convolutional filters in the last layer of the network are divided into several groups, and each group is designed to express specific semantic information under supervision of the filter loss function. The results show which parts of the objects are adopted as the key factors for the network to make the final predicted outcome. In the training process, the network is designed to rapidly focus on the salient feature of objects and played a role in guiding other parts of the network to improve the explainable capability of the network without affecting the discrimination power or compromising the classification accuracy. Extensive experiments based on two public datasets show that the network is highly effective as indicated by high classification accuracy and explainable ability.**

***Index Terms*— Causal inference, explainable visual attention, fine-grained ship classification, remote sensing.**

I. Introduction

MARINE regions cover about two-thirds of the earth and are widely developed and explored. Satellite is an important means of large-scale monitoring in the early stage owing to the wide and complex marine space and is widely applied in studying marine regions. With the improvement of the quantity and quality of remote-sensing images obtained by satellite sensors and the rapid rise of computer vision technology, several methods have made breakthroughs in the field of marine space, such as sea–land segmentation [1], ship detection in optical remote-sensing images [3], [4], ship detection in SAR images [5]–[7], and ship tracking [8]. Resolution of optical remote-sensing images is increased to accurately distinguish the identities of marine objects resulting in the development of fine-grained ship classification task.

Several studies have focused on circumventing the limitation of differentiating ship and non-ship objects [9], [10], which belong to level-1 ship classification tasks. Some studies classify ships into coarse-grained categories, which belong to level-2 ship classification tasks (such as warships or civilian ships) [11], [12]. However, these meta-classification methods and coarse-grained classification methods do not meet the needs of marine monitoring in the real world. An effective fine-grained ship classification framework should be urgently developed owing to the high number of severe maritime security situations. Fine-grained image classification (level-3) should distinguish marginal visual differences with subordinate labels (such as bulk carrier or car carrier), which is more challenging compared with conventional image classification tasks.

Several methods perform well in the field of natural image classification due to advances in deep convolutional networks [13]–[15]. However, it is challenging to transfer this efficacy to the fine-grained ship classification task due to the rich and complex image content of remote-sensing images. In addition, intra-class diversity and inter-class similarity caused by illumination, appearance variation, deformation, and arbitrary orientation complicate classification tasks. A previous study proposed a dual-mask attention network to enhance the ability to distinguish different fine-grained categories and to mine intra-class relationships in dense and complex remote image scenes, thus circumventing these limitations [16]. A multi-level attention mechanism was previously proposed by Qin *et al.* [17] for obtaining discriminative multi-layer features, which helps the network to accurately identify different types of military ships. A previous study designed an end-to-end feature representation network by effectively combining the attribute supervision information and multilevel feature representation to explore the fine-grained information and improve classification performance [18].

In addition to the discrimination power of the network, explainability is a crucial parameter for civil and military applications. Notably, most of the existing fine-grained classification methods based on deep learning utilize statistical learning methods to search for regularities from the massive observed data. These methods train the models to match and fit the training datasets without reasonable explanation for the final decision results. Marine environment is complex and changeable, thus an attention model is required to take a closer look at the differences in key regions and extract discriminative

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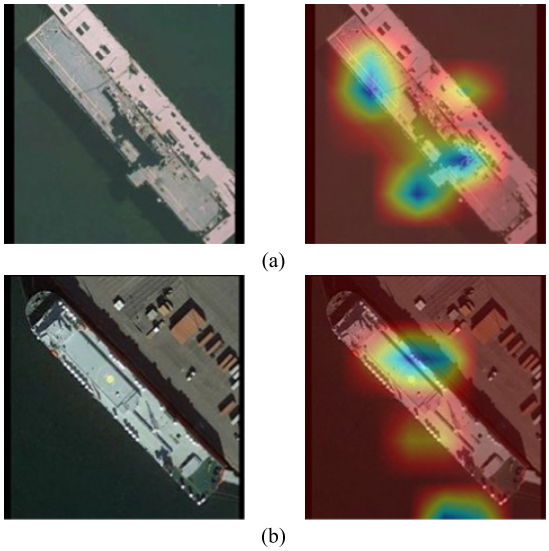


Fig. 1. (a) Image of the amphibious assault ship category from the FGSC-23 dataset and its corresponding activation map. (b) Image of the car carrier category from the FGSC-23 dataset and its corresponding activation map.

features of objects. However, attention models reported in previous studies do not focus on the causality between the attention regions and final predictions. For instance, in the current study, a previously proposed attribute-guided multilevel enhanced feature representation network (AMEFRN) [18] is adopted for fine-grained ship classification on FGSC-23 datasets [18], and gradient-weighted class activation mapping (Grad-CAM) [19] is used for visualization of the features of the correctly classified categories to display how the network affects learning of the features. The final predicted labels are correct, but the visual decision process of the network does not entirely focus on the discriminative region of the objects (see Fig. 1). These spurious associations between attention regions and predicted labels result in unexplainable and unreliable decision-making of the training network. Intrinsic relationships between attention regions of objects and predicted outcomes should be explored to identify true causality from the input images and classification results, thus developing a model that can make decisions in a human logic manner.

In addition to comprehending the model at the logical level, the insufficient explainability at the visual level is a key factor that limits understanding of the existing classification models. Existing deep networks are a black-box model, which only outputs the predicted labels of classification results, but does not provide information on which factors guide the network to make the final prediction. The unexplainable classification results only give the final outcome, but do not provide information on the process. Users can intuitively understand the decision-making process of the classification network at the visual level if the features of the object learned by convolutional filters of the deep network correspond to visual representation that are consistent with human perception, and these visual representations can be further expressed as a linear combination of their contributions to the network's decision.

A novel explainable attention network is proposed in the current study to overcome these limitations. In the causal multi-head attention model (CMAM), a convolutional layer is used to obtain several attention maps from the high-level feature maps to represent the discriminative parts of the objects, thus the network focuses on the salient features of objects. Counterfactual intervention based on causal inference is introduced to explore intrinsic relationships between generated attention maps and predicted categories. Furthermore, in the filter aggregation mechanism (FAM), the filters of the last convolutional layer are divided into several groups to clarify the actual cause behind network decisions at the visual level. Each group represents specific semantic features under the supervision of the filter loss function. As a result, we explicitly present the objects' parts our network focused on and use them for the final prediction. The network can then effectively focus on the multiple salient regions of objects in the early training stage and help to guide other parts of the network in the subsequent stage by dynamically adjusting the weights among the loss functions. Comprehensive experiments based on two public fine-grained ship classification datasets are conducted to validate the discriminative power and explainability of the proposed network.

The main contribution of this article can be summarized as follows.

1) To the best of our knowledge, this is the first work in the field of fine-grained ship classification in remote-sensing images to explore the explainability of a network, whereby the possibility and potential values of the explainable framework in the real-world application are elaborated.
2) In the current study, a novel explainable attention network is proposed for the fine-grained ship classification task. Causal multi-head attention maps are generated to explore the true causality between attention regions and classification results. Filters of the last convolutional layer are divided under the supervision of the filter aggregated loss, to present which regions of the network are used for final prediction. The proposed network helps the users understand decision-making of the network at the logic level and visual level.
3) The proposed network shows powerful classification ability and explainability on two challenging fine-grained ship classification datasets, which demonstrates great potential both in civil and military fields and makes it easy to gain the end-user trust.

The other parts of this article are organized as follows. Section II comprises a review of existing literature related to fine-grained ship classification in remote-sensing images, causal inference, and explainable feature representation. Section III presents the details of the proposed network. Section IV describes the experimental results and analyses. Section V presents the conclusions drawn from this work.

II. Related Work

A. Fine-Grained Ship Classification in Remote-Sensing Images

Limited by the low resolution of optical remote-sensing images, some early studies sought to classify ships and non-ship objects [20]–[22]. The fine-grained ship classification task

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has received high attention during the last few years owing to the advances in image quality and quantity. In a previous study, the first ship classification datasets were generated, namely HRSC2016, which was labeled with bounding-box and three-level categories [11]. Some new baselines have been proposed for different tasks. Shi *et al.* [23] proposed a multi-feature ensemble model to extract the discriminative features by extracting both global and detail features of the ship images. Hypercube-based neuroevolution of augmenting topologies (HyperNEAT) was previously adopted to effectively extract discriminative features of images to overcome the different resolutions and limited data on ship images [24]. A ship detection and category classification framework was recently proposed [28] to classify ship labels from large-scale images. Moreover, a new benchmark, which consists of 23 fine-grained ship categories, was previously proposed [18]. The authors fused the attribute-guided features and fine-grained features to solve the inter-class similarity issue of ship images. A previous study reported a hierarchical multi-label classification method for the classification of ship images in a coarse to fine manner [12]. The authors utilized a hierarchy and exclusion graph and then used different loss functions to supervise the output labels at different levels from two channels. Moreover, an attention model was designed to enhance the ability to distinguish ships in dense and cluttered scenes [16]. A dataset referred to as dataset for oriented ship recognition (DOSR) which comprised 20 ship classes was established to further demonstrate the effectiveness of the attention model. To solve the problem of insufficient training data, a recent study [25] simulated different ship samples and utilized style transfer network to enhance diversity and quantity of the training data. Furthermore, the nearest neighbor prototype based on metric learning was adopted to generate concept feature representation for solving few-shot ship classification [26]. FGSCR-42 dataset was released by a recent study [27] and is the largest public remote-sensing image dataset for fine-grained ship classification.

Most studies only focus on improving the classification accuracy. These methods only output the final predicted labels without explainable reasons. A novel explainable attention network for fine-grained ship classification in remote-sensing images is presented in the current study to clarify the decision-making process.

B. Causal Inference

Explainability has been a major challenge in the development of deep learning. There is an urgent need to open the "black box" of neural networks and design models that make decisions in a human logical manner. Causal inference is a powerful tool for designing the robust and explainable models. It is widely used in the fields of medical science [28], economics [29], and social science [30] in the early stages. However, these methods are designed for the analysis of statistical models with few variables and thus they cannot be effectively applied in the field of computer vision. In "The book of why" [31], the author reported that the existing learning-based models make a prediction by calculating the correlation and statistical distribution on the training datasets. A model does not perform well when the distribution of the testing dataset differs from the training dataset. Most of the human knowledge exists in the form of cause and effect, which does not entirely depend on the observed data. We, humans, are skilled at counterfactual thinking rather than just association by observing, and we wish machine learning can also imitate in this way. Recently, some studies have introduced and used causal inference to the field of computer vision, including zero-shot learning [32], scene graph generation [33], visual dialogue [34], incremental learning [35], attention learning [39], image captioning [36], semantic segmentation [37], and visual question answering [38] to design models that mimic the human logic. In the field of remote sensing, a machine-learning algorithm was proposed in [71] to reconstruct the tracks with poor resolution. This study adopted a causal inference tool to realize the reconstruction of the remote-sensing data. Wang *et al.* [72] proposed a probabilistic graphical model by combining the generative adversarial networks (GANs) and causal inference to capture the underlying relationship from remote-sensing images. Pérez-Suay and Camps-Valls [73] utilized a set of empirical data to estimate the causal relationships among variables and observed data to better understand the governing processes in remote sensing and geosciences. Inspired by these works, the causal intervention is used in the current study to replace the conventional likelihood and eliminate the spurious associations as well as explore the intrinsic relationships between attention regions and classification results.

C. Explainable Feature Representation

Deep neural networks have powerful feature representation capability and thus they are wildly used in several computer vision tasks. Notably, the weakness of deep neural networks is associated with insufficient explainability despite their superior performance [40]. Several methods have been proposed to diagnose the sensitive points of the convolutional layer to explore the black box of convolutional neural networks (CNNs) [41]–[43]. These model diagnosis methods do not exhibit the contribution of the image features to the final decision of the network. Moreover, their explanations are hard to grasp by users.

Visualization of the CNN is the most direct way to clarify the visual meaning of the filters. Previous studies sought to capture the meaning of inner convolutional units [19], [44], [45]. This can be done by searching for maximum activation fields of the hidden layer and visualization of the most important parts of images. The up-convolutional network was proposed previously by Dosovitskiy and Brox [46] to convert image features to images enabling visualization of the feature representation for both conventional hand-designed and learning-based methods. CNNs have the ability to detect objects in the process of scene classification through visualization, thus helping users to understand the feature representations that are learned by the filters of convolutional layers [47].

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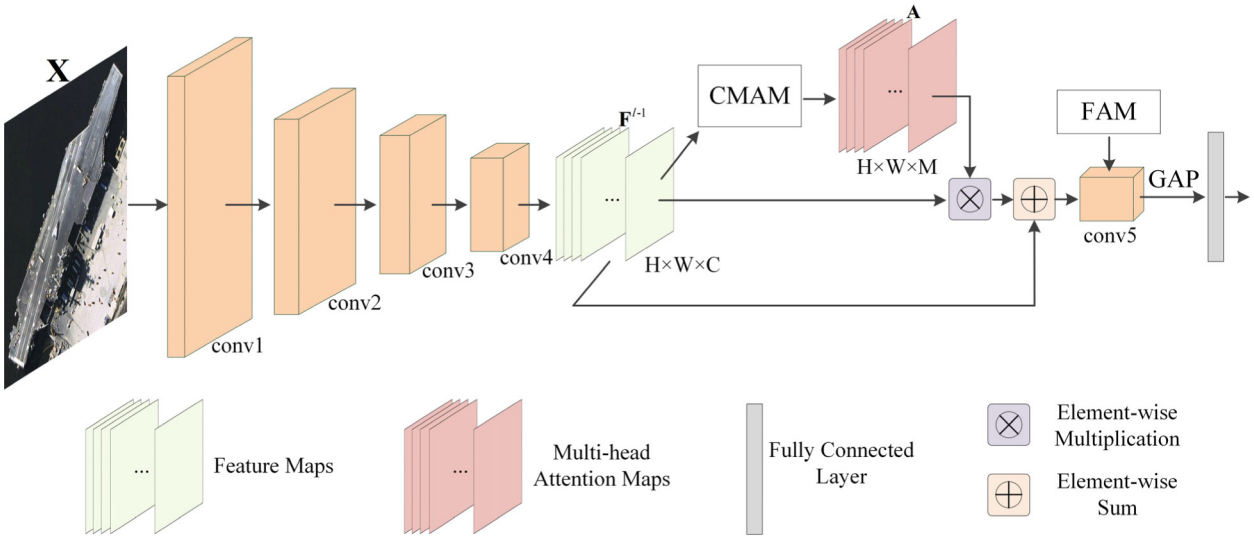


Fig. 2. Schematic representation of the proposed network.

Notably, these methods are post-hoc methods for the trained model. They only provide an approximate understanding and indirect explanation of the original model, which may be inconsistent with the real decision-making behavior of the model, resulting in wrong explanation. Several studies designed explainable CNN models by disentangling the feature representations of convolutional filters to enable convolutional filters to produce more information that are consistent with human-understandable visual concepts [48]–[52]. Inspired by these works, we guide the filters in the last convolutional layer to output the specific attention regions of the ships and help in clarification of the contribution of each part on the final prediction.

III. Proposed Method

A. Overview of the Method

As shown in Fig. 2, we use the deep CNN to extract the high-level features of the input images $\mathbf{X}$. Taking the convolutional backbone VGG16 [53] as an example, the input images are first resized to 224 × 224 before inputting the network. High-level feature maps $\mathbf{F}^{l-1}$ are obtained from Conv4 after a series of convolution and pooling operations.

Furthermore, $\mathbf{F}^{l-1}$ is fed into CMAM to generate multiple causal attention maps $\mathbf{A}$. $\mathbf{F}^{l-1}$ and $\mathbf{A}$ are further fused by multiplying and summing operations to be the inputs of Conv5. Details of CMAM are presented in Section III-B. FAM makes each filter group of the last convolutional layer (Conv5) to output specific regions of objects in images, to provide explainable feature representation. FAM is described in detail in Section III-C.

The total loss function, which comprises filter loss, attention loss, and classification loss, is then used to guide the training process. The loss function is presented in Section III-D.

B. Causal Multi-Head Attention Model (CMAM)

The aim of developing the CMAM model is to explore the true causality between attention maps and predicted labels.

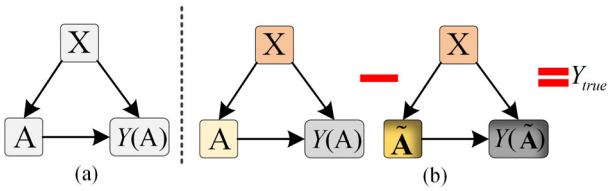


Fig. 3. (a) Causal graph to describe the relationships among the attention maps, input images, and predicted labels. (b) Causal graph for the counterfactual operation.

As shown in Fig. 3(a), we construct the causal graph to show the cause and effect among the attention maps $\mathbf{A}$, input images $\mathbf{X}$, and predicted labels Y.

LinkX $\rightarrow$ $\mathbf{A}$: The images $\mathbf{X}$ are presented as input, and attention maps $\mathbf{A}$ are generated by the multi-head attention model based on CNNs.

LinkX $\rightarrow Y(\mathbf{A}) \leftarrow \mathbf{A}$: The final predicted labels $Y(\mathbf{A})$ are affected by input images $\mathbf{X}$ and attention maps $\mathbf{A}$.

The traditional deep networks operate almost entirely in an associational mode. They are driven by a stream of observed data to which they attempt to fit a function. These models only explore the association but ignore the causality between the observation and final prediction [31]. The network in this study is designed to make decisions that mimic human thinking who have counterfactual thinking, based on results from studies on causal inference. Furthermore, counterfactual causal inference is developed to explore the true effect of attention maps on the predicted labels based on the constructed causal graph [see Fig. 3(a)]. Several false attention maps $\tilde{\mathbf{A}}$ are generated as the counterfactual attention maps and their corresponding predicted labels $Y\left(\tilde{\mathbf{A}}\right)$ are obtained. The true effect of attention regions on predicted results Y_{true} is obtained by calculating the difference between $Y(\mathbf{A})$ and $Y\left(\tilde{\mathbf{A}}\right)$ [see Fig. 3(b)].

Based on the counterfactual operation in Fig. 3(b), we detail the CMAM in Fig. 4. Given an input image $\mathbf{X}$, the high-level features are first extracted by CNNs and the feature maps

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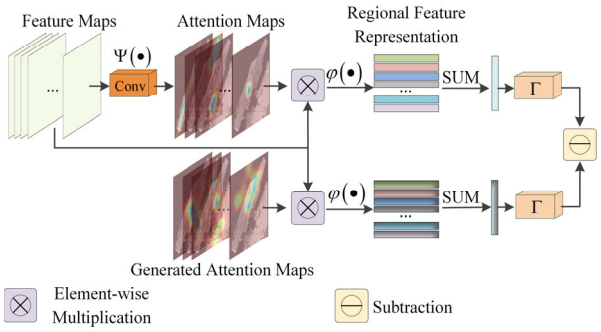


Fig. 4. Framework of the filter aggregation mechanism.

outputting from the Conv4 is represented as $\mathbf{F}^{l-1} \in \mathbb{R}^{W\times H\times C}$, where W, H, and C represent the weight, height, and number of channels of feature maps, respectively. In the first step, the convolutional function is used to obtain M multi-head attention maps $\mathbf{A} = \Psi\left(F^{l-1}\right)$, $\mathbf{A} = \{\mathbf{A_1}, \mathbf{A_2}, \ldots, \mathbf{A}_M\} \in \mathbb{R}^{W\times H\times M}$, where Ψ and M represent the convolutional function with the kernel size of 1×1 and the number of attention maps, respectively. Notably, $\mathbf{A}_m \in \mathbf{A}$ indicates one of the discriminative image regions or visual patterns that the network focuses on. In the second step, the regional feature maps are generated by element-wise multiplying feature maps $\mathbf{F}^{l-1}$ and attention maps $\mathbf{A}$. The global average pooling (GAP) is then adopted to produce the regional feature representation $\mathbf{p} = \{\mathbf{p}_1, \mathbf{p}_2, \ldots, \mathbf{p}_M\}$. It can be expressed as shown in the following equation:

$$\mathbf{p}_m = \varphi\left(\mathbf{F}^{l-1} \odot \mathbf{A}_m\right) \tag{1}$$

where $\odot$ denotes the element-wise multiplication for two tensors. φ represents the GAP operation. The local feature $\mathbf{f}_{\text{local}}$ can be obtained as

$$\mathbf{f}_{\text{local}} = \sum_{m=1}^{M} \mathbf{p}_m = \sum_{m=1}^{M} \varphi\left(\mathbf{F}^{l-1} \odot \mathbf{A}_m\right). \tag{2}$$

A classifier comprising a fully convolutional layer and a batch normalization layer is then designed in this model. We adopt cross-entropy as a loss function, which is denoted as L_{self}. The output of the classifier $\Gamma(\cdot)$ is expressed as follows:

$$Y_{\mathbf{A}}\left(\mathbf{F}^{l-1}\right) = \Gamma(\mathbf{f}_{\text{local}}) = \Gamma\left(\sum_{m=1}^{M} \varphi\left(\mathbf{F}^{l-1} \odot \mathbf{A}_m\right)\right). \tag{3}$$

The essence of the counterfactual based on causal inference is "counter to the facts" [54]. Therefore, the false attention map $\tilde{\mathbf{A}}$ is adopted as the counterfactual attention map. This can be realized by randomly generating some attention maps. The vectors for each location set in counterfactual attention maps follow a uniform distribution with zero mean. The counterfactual local features $\tilde{\mathbf{f}}_{\text{local}}$ can thus be obtained as shown below:

$$\tilde{\mathbf{f}}_{\text{local}} = \sum_{m=1}^{M} \tilde{\mathbf{p}}_m = \sum_{m=1}^{M} \varphi\left(\mathbf{F}^{l-1} \odot \tilde{\mathbf{A}}_m\right). \tag{4}$$

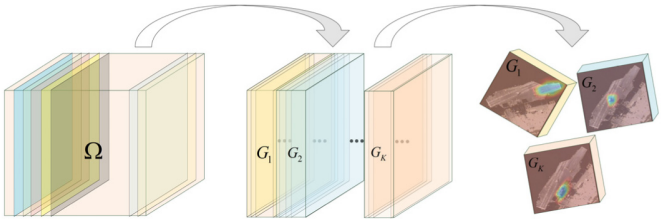


Fig. 5. Framework of the filter aggregation mechanism.

The predicted outcome is calculated according to (3), after the counterfactual operation as shown below:

$$Y_{\tilde{\mathbf{A}}}\left(\mathbf{F}^{l-1}\right) = \Gamma\left(\tilde{\mathbf{f}}_{\text{local}}\right) = \Gamma\left(\sum_{m=1}^{M} \varphi\left(\mathbf{F}^{l-1} \odot \tilde{\mathbf{A}}_m\right)\right). \tag{5}$$

The actual effect of the learned attention on the final prediction can be presented according to the causal graph for the counterfactual operation [see Fig. 3(b)], by calculating the difference between the observed prediction $Y_{\mathbf{A}}\left(\mathbf{F}^{l-1}\right)$ and its counterfactual outcome $Y_{\tilde{\mathbf{A}}}\left(\mathbf{F}^{l-1}\right)$ as shown below:

$$Y_{\text{true}} = Y_{\mathbf{A}}\left(\mathbf{F}^{l-1}\right) - Y_{\tilde{\mathbf{A}}}\left(\mathbf{F}^{l-1}\right) \tag{6}$$

where Y_{true} indicates the true effects of attention maps on the predicted labels. Furthermore, attention loss function Loss_{att} is used in this CMAM to guide the attention learning process and is expressed as follows:

$$\text{Loss}_{\text{att}} = L_{\text{CE}}(Y_{\text{true}}, y) + L_{\text{self}} \tag{7}$$

where y indicates the classification label and L_{CE} represents the cross-entropy loss. This process enables the model to think twice: the first level for prediction from observation L_{self}, and the other for counterfactual imaginary $L_{\text{CE}}(Y_{\text{true}}, y)$.

C. Filter Aggregation Mechanism (FAM)

Based on the causal multi-head attention maps, in this step, we aim to make the convolutional filters to be more explainable in the training process. Each group of filters can present the specific part of the ship through different images of a certain category.

To clarify the contribution of each object's region for the final network decision, we ensure the consistency of the visual patterns of a certain category activated by each group of filters. As shown in Fig. 5, all the filters of the last convolutional layer are presented as $\Omega = \{1, 2, \ldots, d\}$ and are then divided into K groups $G = \{G_1, G_2, \ldots, G_K\}$, where $G_1 \cup G_2 \cup \ldots \cup G_K = G$, $G_1 \cap G_2 = \emptyset$. In the subsequent steps, a filter loss function is added in the last convolutional layer to push the filters in the same groups to focus on a specific object region through different images of a certain category, whereas different groups of filters activate different object's regions.

Attention maps are used as the input of the last convolutional layer and a set of feature maps $\mathbf{F}^l$ is generated as the output. Similarity between the outputs of the filters can be computed as shown below:

$$s_{ij} = C\left(\mathbf{F}_i^l, \mathbf{F}_j^l\right) = \rho_{ij} + 1 = \frac{cov\left(\mathbf{F}_i^l, \mathbf{F}_j^l\right)}{\sigma_i \sigma_j} + 1 \geq 0 \tag{8}$$

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where $\mathbf{F}_i^l$ and $\mathbf{F}_i^l$ indicate the feature maps of the ith filter and the jth filter of the last convolutional layer, respectively. C represents the similarity measure function. ρ_{ij} denotes the Pearson's correlation coefficient (PCC). The following filter loss function is formulated according to the similarity metric

$$\text{Loss}_{\text{filter}} = -\sum_{k=1}^{K} \frac{S_k^{\text{within}}}{S_k^{\text{all}}} \tag{9}$$

where S_k^{within} and S_k^{all} indicate the similarity of intra-group filters and inter-group filters, respectively. The intra-group and inter-group similarities can be obtained as follows:

$$S_k^{\text{within}} = \sum_{i,j \in G_k} s_{ij} = \sum_{i,j \in G_k} C\left(\mathbf{F}_i^l, \mathbf{F}_j^l\right) \tag{10}$$

$$S_k^{\text{all}} = \sum_{i \in G_k, j \in \Omega} s_{ij} = \sum_{i \in G_k, j \in \Omega} C\left(\mathbf{F}_i^l, \mathbf{F}_j^l\right). \tag{11}$$

D. Loss Function

The feature maps of the last convolutional layer are aggregated by the GAP. They are then fed into a fully connected (FC) layer to further classify the features. A cross-entropy loss function is adopted as the classification loss to provide a supervision signal to explicitly guide the explainable attention learning process.

The total loss function of the explainable attention network in the present study is formulated as shown in the following equation, which comprises attention loss Loss_{att}, filter loss $\text{Loss}_{\text{filter}}$, and classification loss $\text{Loss}_{\text{class}}$:

$$\begin{aligned}\text{Loss}_{\text{total}} = &\left(\frac{1}{\text{epoch}+1}\right)\text{Loss}_{\text{att}} \\ &+ \left(1 - \frac{1}{\text{epoch}+1}\right)\text{Loss}_{\text{filter}} \\ &+ \left(1 - \frac{1}{\text{epoch}+1}\right)\lambda\text{Loss}_{\text{class}}.\end{aligned} \tag{12}$$

The proposed network in the current study is supervised by the total loss function and can rapidly focus on the object's regions and guide other parts of the network. This is because the attention loss accounts for a large proportion in the initial training epoch and gradually decrease in the remaining training epoch. Notably, λ presents a hyper-parameter, which is added to balance the importance between $\text{Loss}_{\text{filter}}$ and $\text{Loss}_{\text{class}}$.

IV. Experiments and Analysis

A. Experimental Setup

Two public datasets FGSC-23 [18] and FGSCR-42 [27] are adopted in subsequent experiments to train and evaluate effectiveness of the proposed network. For the FGSC-23 dataset, we followed [18] to randomly select 80% of the images per class for training, whereas the remaining 20% of the images are utilized for testing. Following [27], half of the images of each category from the dataset FGSCR-42 are randomly split for training, while the rest for testing. Considering the number of ships in each class varies significantly for both datasets, a data augmentation strategy is conducted to supplement image samples of some categories in the training set, including image flipping, changing image lights in different levels, random cropping, and rotation. All images are resized to 224 × 224 pixels. Two mainstream convolutional backbones (pre-trained on ImageNet [55]) are adopted to extract the high-level features from the input images, including VGG16 [53] and Restnet50 [56]. Accuracy rate (AR) of each category, overall accuracy (OA), and AU$\left(\overline{\text{PRC}}\right)$ are the three main metrics used to evaluate the classification efficacy of the proposed network. AR determines the ratio of correct predicted labels by the network for a single category on the testing set, OA determines the ratio of the correct predicted labels by the model for all the images on the testing set, whereas AU$\left(\overline{\text{PRC}}\right)$ denotes the area under average precision–recall curve (PRC). All models and datasets are trained for 100 epochs. The stochastic gradient descent (SGD) optimizer with mini-batch is employed for training the model with an initial learning rate set at 0.001. All experiments are implemented based on PyTorch deep learning framework using Ubuntu 16.04, 32 GB of RAM, eight Intel(R) Core(TM) i7-6770K CPU and NVIDIA RTX 2080Ti.

B. Datasets Description

1) FGSC-23: FGSC-23 is a public fine-grained ship classification dataset. It comprises 4080 ship slices and 22 fine-grained categories, including aircraft carrier (AC), destroyer (DE), landing craft (LC), frigate (FR), amphibious transport dock (ATD), cruiser (CR), Tarawa-class amphibious assault ship (TAA), amphibious assault ship (AS), command ship (CS), submarine (SU), medical ship (MS), combat boat (CB), auxiliary ship (AU), container ship (CT), car carrier (CC), hovercraft (HC), bulk carrier (BC), oil tanker (OT), fishing boat (FB), passenger ship (PS), liquefied gas ship (LG), and barge (BA). Each of them is collected from the GF-1 satellite and Google Earth. The image size of these images range from 40 × 40 to 800 × 800. Samples of ship slices from FGSC-23 are presented in Fig. 6. The dataset is available at https://github.com/xiong577/ship-datasets.

2) FGSCR-42: FGSCR-42 is the first large fine-grained ship classification dataset, which has been released in [27] recently. The dataset comprises 9320 images with sizes ranging from 50 × 50 to approximately 500 × 1500. It is divided into 42 fine-grained ship classes including Nimitz class aircraft carrier (NAC), KittyHawk class aircraft carrier (KHCA), Midway class aircraft carrier (MCA), Kuznetsov class aircraft carrier (KNCA), Charles de Gaulle aircraft carrier (CGA), INS Vikramaditya aircraft carrier (IVA), INS Virrat aircraft carrier (IVKA), Ticonderoga class cruiser (TCC), Arleigh Burke class destroyer (ABCD), Akizuki class destroyer (AKCD), Asagiri class destroyer (ASCD), Kidd class destroyer (KDCD), Type 45 destroyer (T45), Wasp class assault ship (WCS), Osumi class landing ship (OCL), Hyuga class helicopter destroyer (HCHD), Lzumo class helicopter destroyer (LCHD), Whitby Island class dock landing ship (WICDL), San Antonio class transport dock (SACTD), Freedom class combat ship (FCC), Independence class combat hip (ICC), Sacramento class support ship (SCS), Crane ship (CS), Abukuma class frigate (ACF), Megayacht (MY), Cargo ship (CGS), Murasame class destroyer (MCD), Container ship (CTS), Towing vessel (TV),

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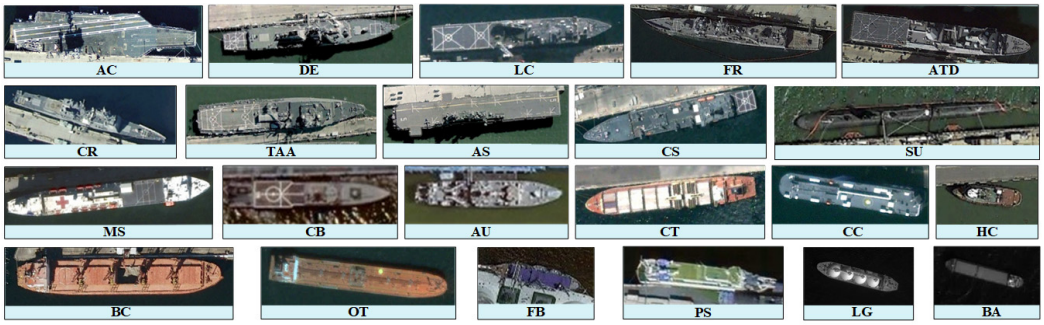


Fig. 6. Examples of ship slices from the FGSC-23 dataset.



Fig. 7. Examples of ship slices from the FGSCR-42 dataset.

TABLE I
EXPERIMENTAL RESULTS OF AR (%) FOR EACH CATEGORIES ON THE FGSC-23 DATASET

Model	AR_1	AR_2	AR_3	AR_4	AR_5	AR_6	AR_7	AR_8	AR_9	AR_{10}	AR_{11}
VGG16	91.26	82.28	91.26	90.29	67.21	82.85	87.05	**87.68**	71.18	88.85	91.11
VGG16+CMAM	**94.08**	**96.20**	**95.01**	**95.99**	**95.82**	**90.02**	**88.10**	87.52	**79.28**	**96.72**	**91.11**
Res50	90.99	86.09	78.13	94.10	84.38	79.98	76.32	83.85	75.18	88.70	99.54
Res50+CMAM	**95.54**	**98.94**	**91.03**	**95.95**	**95.10**	**89.85**	**82.11**	**100.00**	**87.11**	**96.32**	**100.00**
Result	AR_{12}	AR_{13}	AR_{14}	AR_{15}	AR_{16}	AR_{17}	AR_{18}	AR_{19}	AR_{20}	AR_{21}	AR_{22}
VGG16	81.05	78.05	81.32	84.05	100.00	73.25	61.20	61.56	56.87	91.52	**91.85**
VGG16+CMAM	**100.00**	**90.98**	**96.05**	**86.88**	**100.00**	**91.15**	**84.88**	**78.13**	**93.01**	**96.21**	89.91
Res50	94.31	81.22	82.73	91.02	100.00	74.01	73.54	66.87	67.08	100.00	93.85
Res50+CMAM	**95.02**	**85.06**	**92.54**	**93.05**	**100.00**	**90.24**	**78.10**	**76.81**	**89.08**	**100.00**	**100.00**

Civil yacht (CY), Medical ship (MS), Sand carrier (SC), Tank ship (TS), Garibaldi aircraft carrier (GAC), Zumwalt class destroyer (ZCD), Kongo class destroyer (KGCD), Horizon class destroyer (HCD), Atago class destroyer (ACD), Maestrale class frigate (MCAA), Juan Carlos I Strategic Projection Ship (JCSPS), Mistral class amphibious assault ship (MCF), and San Giorgio class transport dock (SGTD). Some ship slice samples from the FGSCR-42 dataset are presented in Fig. 7.

C. Ablation Study

1) Effectiveness of CMAM: An ablation study is conducted on the two datasets to verify the effectiveness of the proposed CMAM. The experimental results of AR for each category on the two datasets are listed in Tables I and II. The findings showed that Resnet50 exhibits better performance for most categories compared with VGG16. The CMAM significantly improves the classification accuracy under the same convolutional backbones. Even if images of some classes exhibit large intra-class variations and small inter-class differences, the proposed method also achieves competitive classification results compared to the backbones. This is because the proposed CMAM helps the network focus on the objects' discriminative parts and explore true causality between attention regions and classification results.

The results of OA and AU are presented in Table III. We can see that the proposed CMAM significantly improves

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TABLE II
EXPERIMENTAL RESULTS OF AR (%) FOR EACH CATEGORY ON THE FGSCR-42 DATASET

Model	VGG16	VGG16+ CMAM	Res50	Res50+ CMAM	Model	VGG16	VGG16+ CMAM	Res50	Res50+ CMAM
AR_1	82.08	**88.42**	86.15	**92.24**	AR_{22}	74.36	**88.03**	82.02	**88.69**
AR_2	79.25	**90.25**	89.15	**92.36**	AR_{23}	76.36	**84.36**	88.02	**98.26**
AR_3	80.25	**91.02**	84.12	**87.42**	AR_{24}	79.87	**86.84**	86.24	**88.96**
AR_4	72.35	**94.85**	92.02	**98.02**	AR_{25}	80.57	**92.65**	82.01	**92.84**
AR_5	84.24	**97.88**	**93.11**	90.36	AR_{26}	40.25	**78.38**	60.12	**96.32**
AR_6	84.02	**92.02**	86.03	**97.82**	AR_{27}	77.02	**90.41**	84.13	**94.26**
AR_7	75.36	**94.68**	88.23	**96.36**	AR_{28}	**80.65**	80.12	85.02	**96.32**
AR_8	80.14	**90.74**	87.02	**92.38**	AR_{29}	42.25	**69.82**	52.56	**94.25**
AR_9	69.26	**96.24**	90.81	**96.28**	AR_{30}	49.36	**66.14**	58.21	**96.98**
AR_{10}	68.12	**88.02**	81.01	**86.04**	AR_{31}	88.25	**90.02**	89.02	**96.25**
AR_{11}	79.25	**96.32**	88.45	**96.38**	AR_{32}	74.25	**79.36**	62.23	**88.75**
AR_{12}	78.25	**89.24**	85.12	**96.25**	AR_{33}	84.25	**89.35**	88.02	**90.25**
AR_{13}	85.26	**98.65**	90.63	**98.98**	AR_{34}	85.36	**88.12**	**90.00**	86.42
AR_{14}	86.26	**94.36**	91.02	**94.02**	AR_{35}	86.36	**98.12**	88.26	**94.26**
AR_{15}	75.36	**88.12**	84.01	**96.28**	AR_{36}	79.32	**86.32**	92.02	**98.64**
AR_{16}	78.25	**92.31**	80.88	**96.02**	AR_{37}	77.69	**94.21**	92.25	**92.35**
AR_{17}	60.23	**94.25**	82.25	**98.26**	AR_{38}	86.02	**92.55**	94.02	**94.25**
AR_{18}	69.25	**86.36**	85.15	**92.36**	AR_{39}	84.36	**90.02**	88.12	**92.26**
AR_{19}	78.25	**96.32**	87.44	**94.02**	AR_{40}	82.02	**84.36**	80.02	**96.88**
AR_{20}	84.25	**94.23**	95.87	**96.34**	AR_{41}	79.02	**84.25**	89.28	**96.08**
AR_{21}	90.02	**92.02**	91.02	**94.25**	AR_{42}	75.21	**94.15**	90.12	**94.26**

TABLE III
RESULTS OF OA (%) FOR THE TWO DATASETS

Dataset	Model	OA	AU
FGSC-23	**VGG16**	81.43	91.34
	VGG16+CMAM	**91.14**	**96.03**
	Res50	84.05	91.86
	Res50+CMAM	**92.02**	**96.21**
FGSCR-42	**VGG16**	76.71	87.95
	VGG16+CMAM	**89.38**	**94.68**
	Res50	84.77	91.19
	Res50+CMAM	**94.06**	**97.63**

the classification accuracy for different convolutional backbones on the two datasets with two evaluation metrics. For the FGSC-23 dataset, the proposed CMAM improves the OA by 9.71% and 7.97% with VGG16 and Resnet50, respectively. Our method also improves the AU by 4.69% and 3.35% with VGG16 and Resnet50, respectively. And for the FGSCR-42 dataset, the proposed CMAM can offer 12.67% and 9.29% performance improvement for VGG16 and Resnet50 with OA metric, respectively. Our method can also offer 6.73% and 6.44% performance improvement for VGG16 and Resnet50 with AU metric, respectively. The results of two evaluation metrics improve with the deepening of the convolutional backbone for the two datasets. In addition, improvement of results with CMAM on the FGSCR-42 dataset is significantly higher compared with that of the FGSC-23 dataset. This is mainly attributed to the characteristics of large intra-class diversity and small inter-class similarity which are significant in FGSCR-42 relative to those in the FGSC-23 dataset.

Gradient-weighted class activation mapping (Grad-CAM) [19] is applied on the two datasets to visualize how the network affects the learning of the class-specific features to intuitively show the discriminative and reliability of the attention features used in training of CMAM. The top convolutional layers of the Resnet50 and Res50+CMAM are chosen for visualization. As shown in Fig. 8, the first two rows indicate the heatmaps of the images from FGSC-23 dataset, whereas the last two rows are the experimental results of the FGSCR-42 dataset. The proposed Res50+CMAM helps the network to capture more discriminative regions of the ship objects and to ignore other irrelevant parts of the images compared with conventional convolutional backbone Resnet50.

2) Effectiveness of FAM: Activation parts of filter groups of the last convolutional layer are visualized to verify the effectiveness of the proposed FAM. As shown in Fig. 9, the first three rows indicate visualization results of the images

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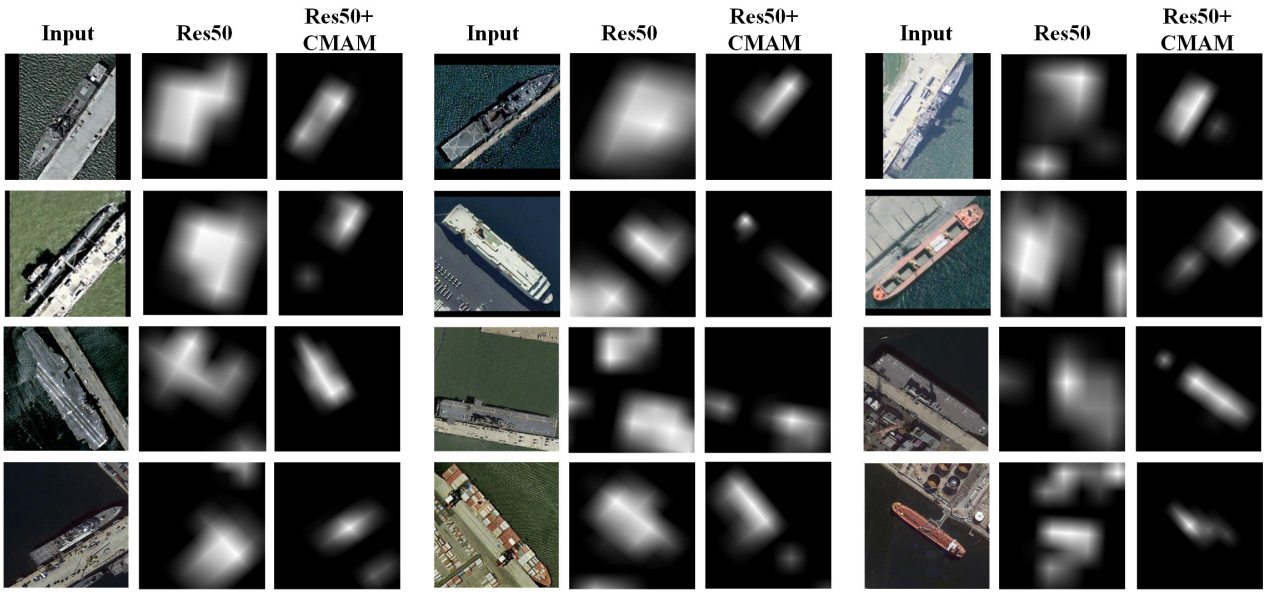


Fig. 8. Examples of attention maps on two datasets. The first two rows present heatmaps of images from the FGSC-23 dataset, whereas the last two rows present the experimental results of the FGSCR-42 dataset.

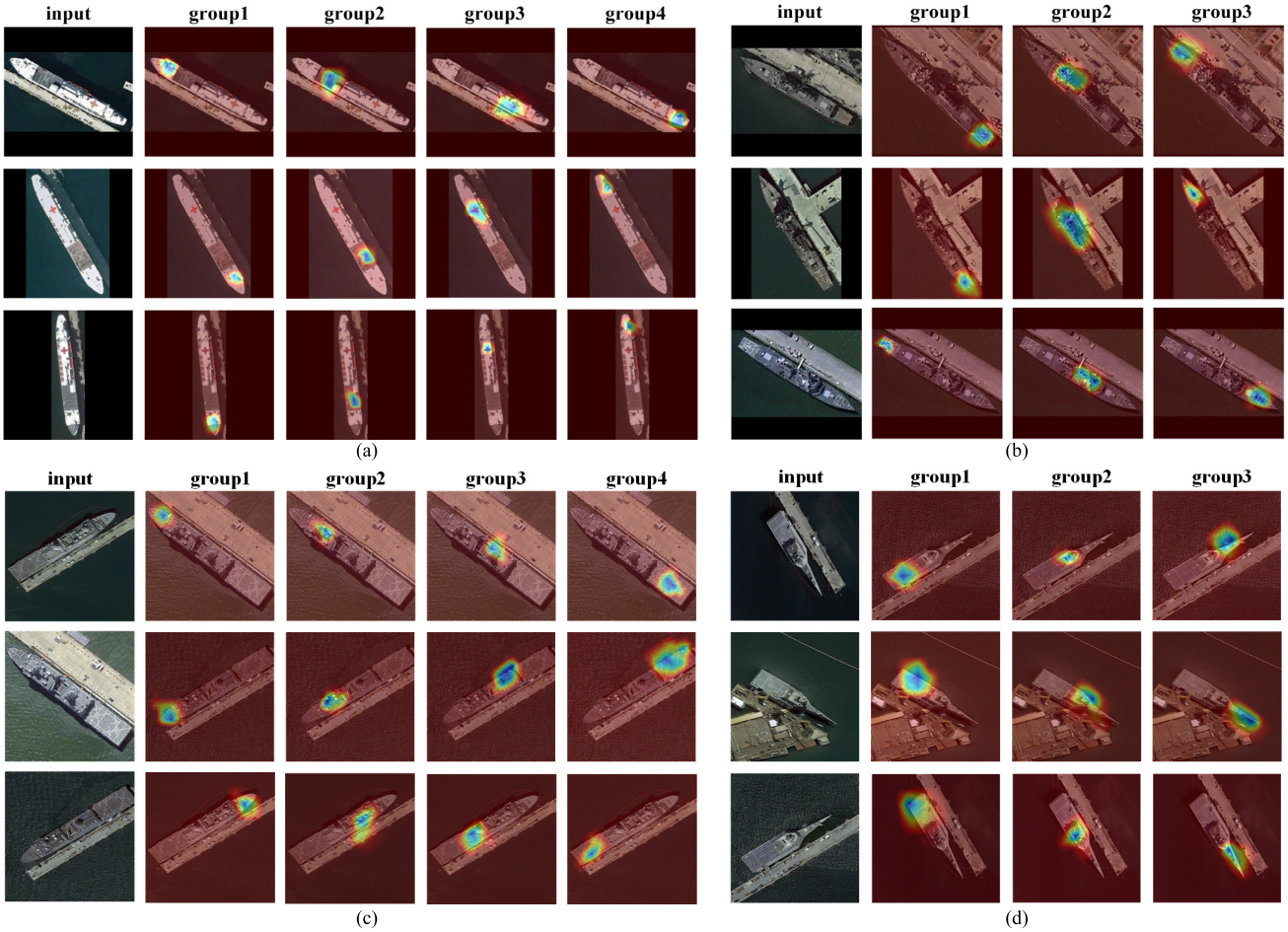
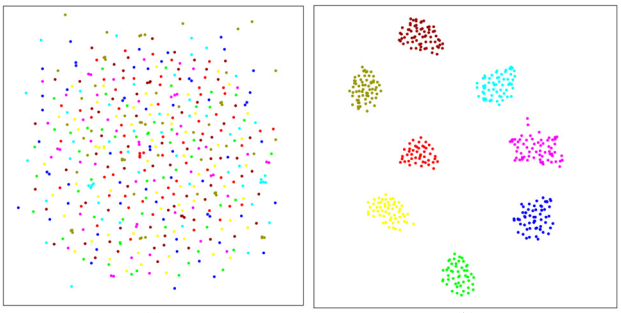


Fig. 9. Visualization results of feature maps for some groups of filters. The first three rows are visualization results of the images from FGSC-23 [(a) and (b)], while the last three rows are the experimental results of FGSCR-42 [(c) and (d)].

from the FGSC-23 dataset, whereas the last three rows present the experimental results of the FGSCR-42 dataset. Images in each column excluding the input images represent the activated regions in the feature maps from the same group of filters. Analysis of different images of the same category shows that each group of filters are activated by a certain region

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(a) (b)

Fig. 10. Feature distribution of learned features for some groups of filters on the FGSCR-42 dataset. (a) Feature visualization of Res50+CMAM. (b) Feature visualization of Res50+CMAM+FAM.

TABLE IV
RESULTS OF OA (%) ON TWO DATASETS

Dataset	Model	OA
FGSC-23	VGG16+CMAM	91.14
	VGG16+CMAM+FAM	91.08
	Res50+CMAM	92.02
	Res50+CMAM+FAM	91.78
FGSCR-42	VGG16+CMAM	89.38
	VGG16+CMAM+FAM	89.19
	Res50+CMAM	94.06
	Res50+CMAM+FAM	94.06

of the objects. The same group of filters represent similar specific visual information of the objects, whereas different groups represent diverse visual information of the objects. For instance, in Fig. 9(b), groups 1–3 indicate that the afterdeck region, superstructure/midship region, and foredeck region are activated, respectively. Confusion feature representation of conventional CNN models significantly hinders understanding of the decision-making process and limits application of the models. The proposed FAM provides meaningful visual information for each group of filters, thus helping users understand the regions that the network pay attention to when classifying ships, and wins the trust of users. Moreover, the clear feature representation can help in quantifying the contribution of different object regions to the final predictions in the subsequent experiments.

The t-distributed stochastic neighbor embedding (t-SNE) algorithm [57] is utilized to obtain the 2-D feature representations of the two approaches, to further show the differences in features represented by the traditional convolutional backbone (Res50+CMAM) and FAM (Res50+CMAM+FAM). As shown in Fig. 10, the feature visualization of Res50+CMAM displays the chaotic distribution. However, the feature distribution of Res50+CMAM+FAM is more compact relative to the feature distribution in Res50+CMAM. This is largely attributable to the proposed FAM, which leads to separation in different semantic information and pulls the kernels that output similar semantic information closer together in the feature space.

In addition to clarifying the object's regions on which the network focuses during decision-making process, it is necessary to quantify each part and show the contribution of each attention part to the final prediction for users. As shown in Fig. 11, the methods described in [52] is used to obtain quantization results. The first row shows the quantization results of images obtained from FGSC-23, while the last row displays the quantization results of images derived from FGSCR-42. Some pie charts are constructed to intuitively show the contribution of each region of the object in images. All the quantization results provide clear and correct explanations of the network for the users.

To validate the effectiveness and feasibility of the proposed FAM, the classification accuracy is tested after introducing the proposed FAM. The results of OA are shown in Table IV. The proposed network (VGG16/Res50+CMAM+FAM) yields comparable results on two datasets, which hurt the classification accuracy less than 0.5%. Therefore, the proposed network provides clear explainable classification results and controls the reduction of classification accuracy within an acceptable range.

D. Parameter Analysis

The hyper-parameters M, K, and λ are important in the proposed network. In this section, several experiments are carried out to analyze the effects of these parameters on the performance of the proposed network. Table V and Fig. 12(a) and (d) show the effects of the number of attention heads (M) for classification accuracy. For the FGSC-23 dataset, M is set to 16 and 32 for VGG16 and Resnet50, respectively, to produce the best classification results for the network. For the FGSCR-42 dataset, the best choice of the attention number is set to 32 for both VGG16 and Resnet50. The effects of the number of convolutional groups (K) are illustrated in Table VI and Fig. 12(f) and (e). The results show that the proposed network is not sensitive to the parameter K. With the increase of the number of convolutional filters, there is no significant change in classification accuracy. Table VII and Fig. 12(c) and (f) demonstrate that the proposed network achieves the best performance when λ is set to 0.01 for VGG16 and 0.1 for Resnet50.

E. Comparison With Other Methods

To compare the performance of the proposed network with existing ones, the classification accuracy of the proposed method is compared with that of other methods on two datasets, as shown in Table VIII. All the methods shown are based on deep learning-based features. For the FGSC-23 dataset, Inception-v3 [58], DenseNet121 [59], MobileNet [60], and Xception [61] are the most commonly used CNN models. Xception combines ResNet and Inception series to yield better performance than the other three deep CNN models. FDN [62], ME-CNN [63], LGFFE [64], and AMEFRN (sc2) [18] have been proposed for remote-sensing image classification tasks. By fusing global and local image information, AMEFRN (sc2) outperforms the other three methods. B-CNN [65] and DCN [66] have been proposed for fine-grained natural image

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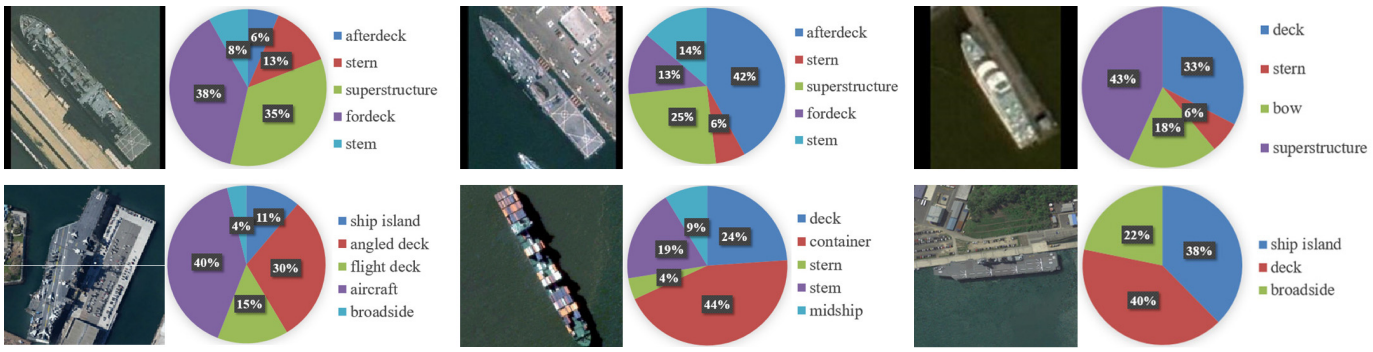


Fig. 11. Contribution of each attention part to the final predictions. The first row is quantization results of the images from FGSC-23, while the last row is the quantization results of FGSCR-42.

TABLE V

COMPARISONS OF CLASSIFICATION ACCURACIES UNDER DIFFERENT NUMBERS OF ATTENTION HEADS (M) ON THE TWO DATASETS

Dataset	Model	M=1	M=4	M=8	M=16	M=32	M=64
FGSC-23	**Vgg16**	89.04	90.68	90.81	**91.08**	91.01	90.14
	Res50	90.04	90.13	91.48	91.54	**91.78**	91.14
FGSCR-42	**Vgg16**	87.12	88.36	88.81	89.07	**89.19**	88.03
	Res50	90.01	92.13	93.51	93.85	**94.06**	93.67

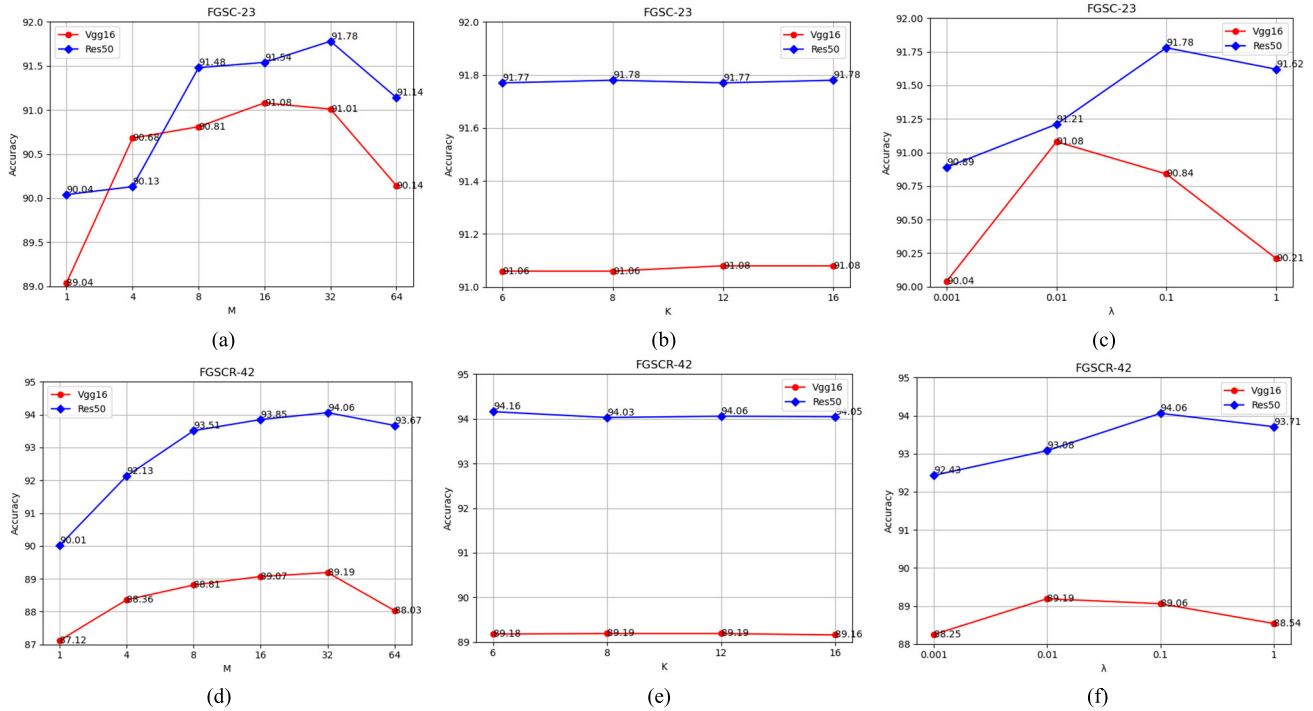


Fig. 12. Curves of classification accuracy with respect to different parameters under the two datasets.

recognition tasks, and B-CNN [65] is inferior to DCN [66]. Likewise, for the FGSCR-42 dataset, general CNN models are used to make comparisons, including VGG19 [53], ResNet-50 [56], DenseNet [59], and ResNext-50 [67]. Inspired by ResNet and Inception, ResNext-50 improves nonlinear ability of the model and outperforms the other general models. B-CNN [65], RA-CNN [68], DCL [69], and TASN [70] are four fine-grained image recognition models. TASN captures important details of the whole images using the proposed trilinear attention module and achieves better classification results compared to other methods. The proposed network achieves the best classification results on both FGSC-23 and

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TABLE VI
COMPARISONS OF CLASSIFICATION ACCURACIES IN DIFFERENT NUMBERS OF FILTER GROUPS (K) ON THE TWO DATASETS

Dataset	Model	K=6	K=8	K=12	K=16
FGSC-23	**Vgg16**	91.06	91.06	**91.08**	91.08
	Res50	91.77	**91.78**	91.77	91.78
FGSCR-42	**Vgg16**	89.18	89.19	**89.19**	89.16
	Res50	94.06	94.03	**94.06**	94.05

TABLE VII
COMPARISONS OF CLASSIFICATION ACCURACIES UNDER DIFFERENT VALUES OF PARAMETER (λ) ON THE TWO DATASETS

Dataset	Model	λ =0.001	λ =0.01	λ =0.1	λ =1
FGSC-23	**Vgg16**	90.04	**91.08**	90.84	90.21
	Res50	90.89	91.21	**91.78**	91.62
FGSCR-42	**Vgg16**	88.25	**89.19**	89.06	88.54
	Res50	92.43	93.08	**94.06**	93.71

TABLE VIII
COMPARISON OF CLASSIFICATION ACCURACY OF DIFFERENT METHODS

Dataset	Model	OA
FGSC-23	Inception-v3 [58]	83.88
	DenseNet121 [59]	84.00
	MobileNet [60]	84.24
	Xception [62]	87.76
	FDN [62]	82.30
	ME-CNN [63]	85.58
	LGFFE[64]	89.45
	AMEFRN(sc2) [18]	91.15
	B-CNN [65]	84.00
	DCN [66]	90.66
	ours	**91.78**
FGSCR-42	VGG19 [53]	77.36
	ResNet-50 [67]	84.77
	DenseNet [59]	88.69
	ResNext-50 [18]	89.16
	B-CNN [65]	89.53
	RA-CNN [68]	91.63
	DCL [69]	93.03
	TASN [70]	93.51
	ours	**94.06**

FGSCR-42 datasets. This further shows that the proposed network is not only able to provide feature explanation but have powerful discrimination.

V. CONCLUSION

In this article, a novel explainable attention network is proposed for fine-grained ship classification in remote-sensing images. The network mainly consists of CMAM, which extracts multiple causal attention maps from the input images and FAM, thereby facilitating the generation of explainable image features. For the CMAM, a convolutional layer is used to obtain multiple attention maps from high-level feature maps. A causal graph is constructed through the counterfactual strategy and used to explore the intrinsic relationships between the attention region and predicted outcomes. For the FAM, a filter loss helps the filter groups to represent the specific regions of objects in images and clarify the contribution to final prediction. Extensive experiments are conducted on two public fine-grained ship classification datasets. The proposed method achieves the best results compared with the existing methods and provide explainable feature visualization. Meanwhile, with the rapid development of remote-sensing technologies, the satellite can capture ship images from different modalities. In our future work, we will extend our explainable network to multi-modality remote-sensing ship image classification tasks.

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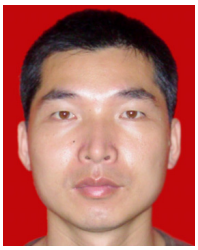
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